CYBERSECURITY AWARENESS WORKSHEET

Spot the Secure Website: HTTP vs HTTPS

Worksheet completed by Oliva

This activity helps participants identify secure vs insecure websites by analysing URL structure, browser indicators, and trust signals.

The goal is to improve awareness of how to recognise potentially unsafe websites and understand that HTTPS alone does not guarantee legitimacy.

Section 1: Website Security Evaluation

Website	URL	Indicators	Secure? (Yes/No)
A	http://online-banking-login.example.com	No padlock, HTTP	No
B	https://university.edu/login	Padlock, HTTPS	Yes
C	https://secure-paypal-login.example.net	HTTPS but suspicious domain	No

Questions:

Which website is most secure? Why?

B, as it has the padlock indicator enabled.

Why is HTTP considered unsafe?

People can steal your data over the network.

Does HTTPS always mean a site is trustworthy? Explain.

Secure communication, but website may be unsafe.

Section 2: Safe vs Unsafe Links

Evaluate the following links-

Task:

- Mark each link as Safe / Suspicious
- Identify one red flag for each suspicious link

1. https://paypal.com/login

Safe – Looks standard

2. https://paypal-secure-login.verify-now.example.com

Suspicious – Long URL

3. https://uwa.edu.au

Safe – Looks normal

4. http://free-prizes.example.net

Suspicious – Free prizes look dodgy

Section 3: File Safety (Malware Awareness)

Determine whether these files are safe:

- invoice.pdf.exe
- report.docx
- update_patch_v2.zip
- photo.png

Questions:

- Which file is most suspicious? Why?
- What is a common trick used in malicious file names?

Invoice.pdf.exe – Suspicious, as it has double file extension

Report.docx – Safe, doesn't look suspicious

Update_patch_v2.zip – Suspicious, as it could contain hacker code

Photo.png – Safe, Looks like normal photo

Most suspicious file is first one.
Double extension is a common trick used in malicious file names.

Section 4: Password Strength Analysis

Rank the following passwords from weakest → strongest:

☐1 123456

☐2 password2024

☐3 sam12345

☐4 UWA!secure9

☐5 Xy9#kL2pQ

Questions:

- What makes a password strong?
- Why is password reuse dangerous?

Strong passwords have random symbols, numbers, and letters.

Password reuse is dangerous, allows hackers to get in to different accounts.

Section 5: Social Engineering Scenarios

Scenario 1:

You receive a message from "IT Support" asking for your password to fix an issue.

Scenario 2:

A friend messages you urgently asking for money.

Questions:

- What are the warning signs?
- What would you do in each situation?

Scenario 1 – Asking for passwords is a warning sign.
Could ask IT office.

Scenario 2 – Asking for money might be dodgy.
Talk to them.

Section 6: Quick Knowledge Check (Mixed Questions)

Multiple Choice:

1. What does HTTPS provide?
   ☐ Encryption
   ☐ Faster internet speed
   ☐ Virus protection

2. What is phishing?
   ☐ A type of malware
   ☐ A social engineering attack
   ☐ A firewall

True / False:

- HTTPS guarantees a website is completely safe (T / F) - False
- Strong passwords should include symbols and numbers (T / F) - True

Section 4: Reflection

What new cybersecurity concept did you learn?

Password security, and different ways of strengthening them.

Which section did you find most difficult?

Malicious file analysis.

How will this activity change your online behaviour?

Don’t trust all website until further analysis.